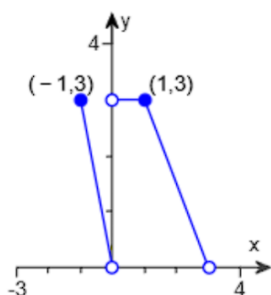


Question 1: Find the Domain and Range of $G(t) = \frac{1}{t^2-81}$

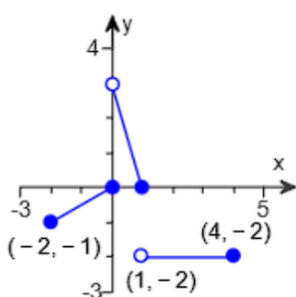
Question 2: Find the natural domain of $F(t) = \frac{5t}{|t|}$ and its graph

Question 3: Find a formula for each following graphed function
With full dot means $[$, blank dot means $($)

a.



b.



Question 4: Determine algebraically whether the given function is even, odd or neither

$$h(x) = \frac{-9x^3}{8x^2-9}$$

Question 5: Graphs the functions $f(x) = \frac{x}{5}$ and $g(x) = 1 + \frac{10}{x}$ together. What are the values of x for which $\frac{x}{5} > 1 + \frac{10}{x}$? When the given inequality is rewritten so that there is a 0 on the right side and the left side is a single rational expression, the result is _____. The numerator is 0 if $x =$ _____ and the denominator is 0 if $x =$ _____. These values divide the real line into intervals. Using a test value from each interval confirms that $\frac{x}{5} > 1 + \frac{10}{x}$ for values of x on _____.

Question 6: $f(x) = \sqrt{x+6}$, $g(x) = \frac{1}{x}$. The formula for the composite function $f \circ g$ is _____. The formula for the composite function $g \circ f$ is _____. The domain of $f \circ g$ is _____. The domain of $g \circ f$ is _____.

Question 7: Find the domain and range for the function $f(x) = \frac{3}{1-e^{5x}}$

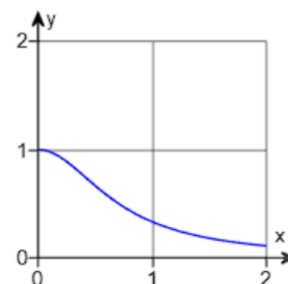
Question 8: Graph the following function and determine whether the function is 1-to-1:

$$f(x) = 1 - \frac{x}{4} \text{ if } x \leq 0 \text{ and } f(x) = \frac{x}{x+4} \text{ if } x > 0$$

Question 9: The graph of the function $f(x) = \frac{1}{2x^2+1}, x \geq 0$, is shown.

Use symmetry with respect to the line $y = x$ to add the graph of $f^{-1}(x)$ on the same graph (It is not necessary to find a formula for $f^{-1}(x)$).

Identify the domain and range of $f^{-1}(x)$



Question 10: Given $f(x) = \frac{x+4}{x-5}$. Find $f^{-1}(x)$. What is the domain of $f^{-1}(x)$? What is the range? Does $f(f^{-1}(x)) = f^{-1}(f(x)) = x$? Find the intermediate forms of $f(f^{-1}(x))$ and $f^{-1}(f(x))$